

DIGITAL ELECTRONICS

- ① Boolean algebra deals with zeroes & ones only but not with don't cares
K-map deals with zeroes, ones & don't cares

② $(A+B)(A+C) = A+BC$

③ we have, $(A+B)(A+C) = A+BC$

$$A+BC = (A+B)(A+C)$$

||| $A+\bar{A}B = (A+\bar{A})(A+B) = A+B$

$$\therefore A+\bar{A}\bar{B} = A+\bar{B}$$

$$\bar{A}+AB = \bar{A}+B$$

$$\bar{A}+A\bar{B} = \bar{A}+\bar{B}$$

④ Demorgan's Theorem

It can be used to find out the compliment of a given expr.

$$f'(A, B, C, 0, 1, +, \cdot) = f(\bar{A}, \bar{B}, \bar{C}, 1, 0, \cdot, +)$$

① $\overline{AB} = \bar{A} + \bar{B}$

② $\overline{A+B} = \bar{A} \cdot \bar{B}$

⑤ * Principle of Duality

It can be used to convert a positive logic system into a negative logic system & viceversa.

$$f^D(A, B, C, +, \cdot, 1, 0) = f(A, B, C, \cdot, +, 0, 1)$$

Eg: If $A+\bar{A}=1$ then $A \cdot \bar{A}=0$

Self dual expressions: If we perform the dual operation once & still we'll get back the original expression.

⑥ Consensus Theorem

If any exp is in the form of $AP_1 + \bar{A}P_2 + P_1P_2$ then the term P_1P_2 is redundant

$$\text{i.e. } AP_1 + \bar{A}P_2 + P_1P_2 = AP_1 + \bar{A}P_2$$

⑦

Variables	input combinations	logical expressions
n	2^n	2^{2^n}
2 ($\bar{A}B, B$)	4 00 01 10 11	16 $AB \quad A+B \quad 0 \quad \bar{A}$ $A\bar{B} \quad A+\bar{B} \quad 1 \quad \bar{B}$ $\bar{A}B \quad \bar{A}+B \quad A \quad A\bar{B}+\bar{A}B$ $\bar{A}\bar{B} \quad \bar{A}+\bar{B} \quad B \quad AB+\bar{A}\bar{B}$

⑧ Find out equivalent -ve logic system for the AND gate
* in the +ve logic system

Use Duality Theorem

$$\text{i.e. } f(A,B) = AB$$

$$f^D(A,B) = A+B \quad \text{i.e. OR gate.}$$

⑨ SOP exp for a given Truth Table is equal to POS exp but not always

If the Truth Table contains any don't cares, then POS need not be equivalent to SOP.

⑩

Canonical SOP

If each term of an SOP term contains all the literals then it is said to be canonical SOP exp

$$\text{Eg: } f(A,B,C) = ABC + \bar{A}\bar{B}C$$

⑪ Each Term in canonical SOP is called as min-Term
 —||— || —||— POS —||— max-Term.

⑫ * Find out the no of min Terms in $f(A,B,C) = A+BC$

A + BC			
↙		↓	
A	(B)	(C)	
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

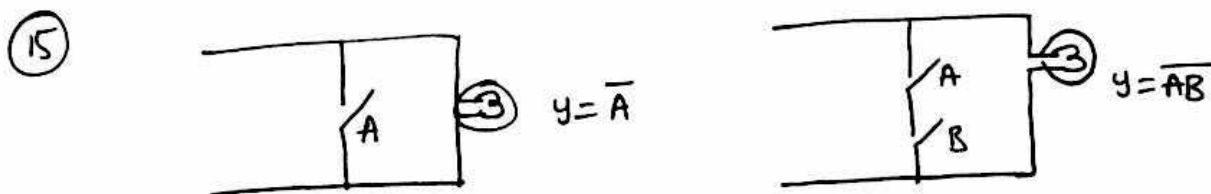
(A)	B	C	
0	1	1	m_3
1	1	1	m_7 (Repeated)

∴ 5 min Terms.

A	B	C	minTerm	maxTerm
0	0	1	$\bar{A}\bar{B}C$	$A+B+\bar{C}$
0	1	1	$\bar{A}BC$	$A+\bar{B}+\bar{C}$

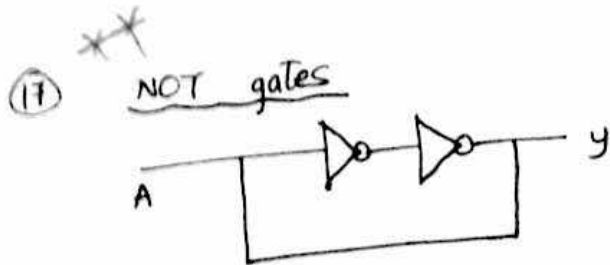
} minTerm = maxTerm

⑬ SOP exp, $f(A,B) = \sum m(0,1,2,3)$
 POS exp, $f(A,B) = \prod M(0,1,2,3)$ } Representation

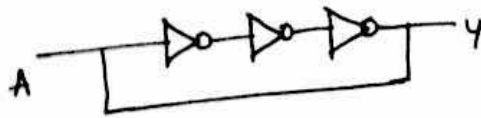


⑯ In K-map

pair	eliminates	1	variable
Quad	— —	2	— —
Octet	— —	3	— —



Bi-stable multi vibrator



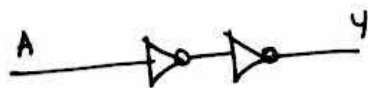
Astable multi vibrator



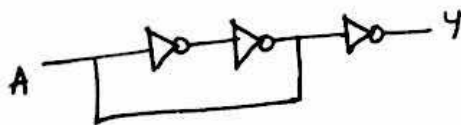
$$T_{ON} = T_{OFF}$$

$$\therefore T = 2T_{ON} = 2 \times \text{PD of NOT gates}$$

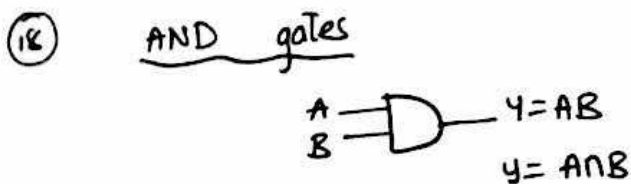
$$* f = \frac{1}{2T_{ON}} = \frac{1}{T}$$



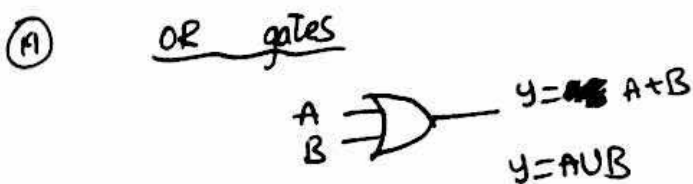
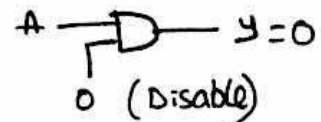
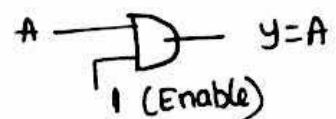
Buffer (no feed back)



Bi-stable multi vibrator.



Enable / Disable ckt

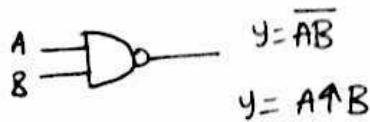


Enable is 0

Disable is 1

⑭ * Floated inputs in TTL logic family can act as logic 1 whereas in ECL family it will act as logic 0.

(21) NAND gates



$$A \uparrow 0 = 1$$

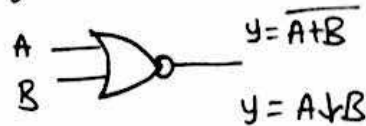
$$\left. \begin{array}{l} A \uparrow 1 = \bar{A} \\ A \uparrow A = \bar{A} \end{array} \right\} \text{Inverter logic}$$

$$A \uparrow \bar{A} = 1$$

Enable = logic 1

Disable = logic 0

(22) NOR gates



$$A \downarrow 0 = \bar{A}$$

$$A \downarrow 1 = 0$$

$$A \downarrow A = \bar{A}$$

$$A \downarrow \bar{A} = 0$$

Inverter logic

Enable = logic 0

Disable = logic 1

(23) * E-XOR gates

odd function

Inequality detector

odd parity checker

even parity generator

$$A \oplus 1 = \bar{A} \quad \{ \text{Inverter logic} \}$$

* Both 0 & 1 are enable i/ps.

(24) * EX-NOR gates

Even func

Equivalence gate

Even parity checker

odd parity generator

$$A \odot 0 = \bar{A} \quad \{ \text{Inverter logic} \}$$

Both 0 & 1 are enable i/ps

(25) * Functionally complete set

Any boolean exp can be implemented by using the logic gates available in that set alone.

(26) * Equivalent symbols

XOR



≡



$$Y = A \oplus B$$

X-NOR



≡



$$Y = A \odot B$$

(27) The complete set of only those logic gates designated as universal gates are NAND & NOR

(28) Convert $(0.25)_{10}$ into its equivalent binary

$$0.25 \times 2 = 0.5 \quad \text{---} \quad 0$$

$$0.5 \times 2 = 1.00 \quad \text{---} \quad 1$$

$$\therefore (0.25)_{10} = (0.01)_2$$

(29) MUX

Many To one circuit

Data selector

parallel to Serial converter

Universal logic circuit

(30)* Mux can be used to realize SOP exp only

(31)* To design an $N \times 1$ mux by using 2×1 mux, we require $(N-1)$ no of 2×1 muxs

(32)* Min no of 4×1 muxes required to design 32×1 mux

$$\frac{32}{4} \text{ (8)} + \frac{8}{4} \text{ (2)} + \frac{2}{4} \text{ (0.5)} = 10.5 \approx 11 \text{ muxes.}$$

(33) De-mux

One To many circuit

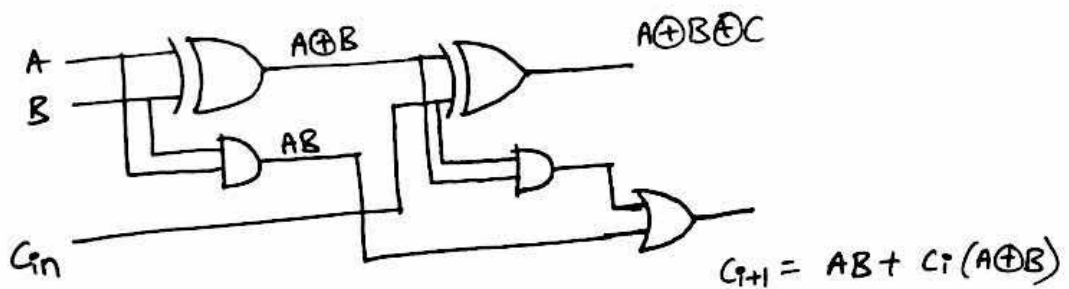
Data distributor

cannot be called as serial to parallel converter

(34)

CIRCUITS	min no of NAND/ NOR gates		propagation delay
	NAND	NOR	
Half adder	5	5	1 logic gate delay
Half subtractor	5	5	1 logic gate delay
Full adder	9	9	2 logic gate delay
Full subtractor	9	9	2 logic gate delay

(35)

Carry look ahead adder

$$P_i = C_p = A \oplus B$$

$$G_i = C_g = AB$$

$$C_{i+1} = G_i + C_i P_i$$

$$\therefore C_1 = G_0 + C_0 P_0$$

$$C_2 = G_1 + C_1 P_1 = G_1 + (G_0 + C_0 P_0) P_1$$

$$= G_1 + G_0 P_1 + C_0 P_0 P_1$$

(36)

	AND array	OR array
ROM	F	F
PROM	F	P
PAL	P	F
PLA	P	P

F → Fixed

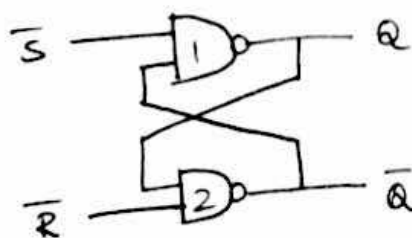
P → Programmable.

37) PROM { Programmable Read only memory }

The memory capacity can be repr by $2^k \times n$

where, $k \rightarrow$ No of i/p bits
 $n \rightarrow$ No of o/p bits.

38) * $\overline{S}\overline{R}$ Nand latch

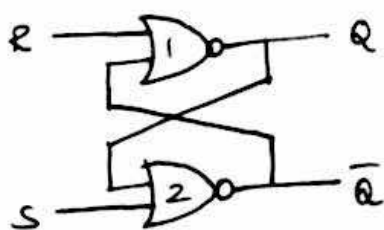


1 & 2 works @ diff speed
 can't say which is faster

$\overline{S}$	$\overline{R}$	o/p (Q_{n+1})
0	0	Invalid
0	1	1
1	0	0
1	1	Q_n

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Invalid $Q = \overline{Q} = 1$

NOR latch



S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$Q = \overline{Q} = 0$ (Invalid)

39) * The condition to occur race around is

$T_{pff} < T_p$ where, $T_{pff} \rightarrow$ Prop delay of FF

$T_p \rightarrow$ ON Time period of a clock

condition to avoid race around is

$$T_p < T_{pff} < T$$

where, $T \rightarrow$ Period of clock.

Race around can be eliminated by

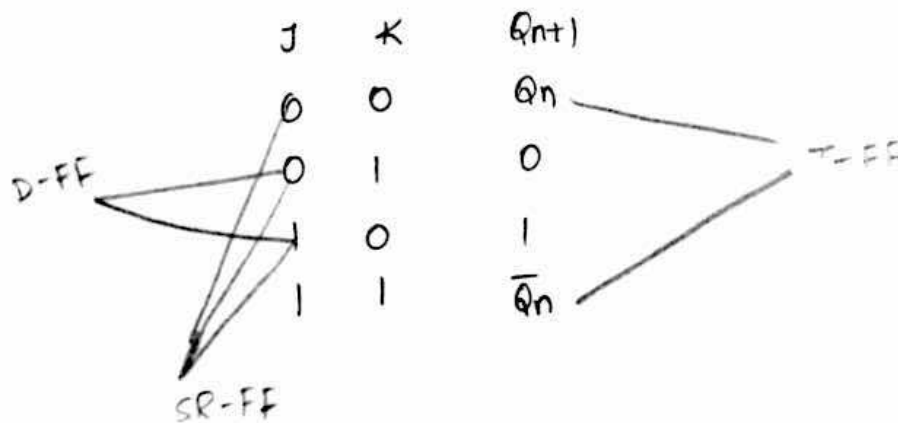
(a) master-slave configuration

(b) Edge Triggering

40 * JK master slave configuration

- (a) master is level sensitive whereas slave is edge sensitive
 (b) Both master & slave are SR-FFs only.

41 JK-FF can be called as universal FF



42 (a) If T_{pd} be the Time delay of the flipflop, then Time delay offered by ripple counter is $N \times T_{pd}$ where $N \rightarrow$ No of o/p bits.

(b) Time delay of a synchronous counter is T_{pd} irrespective of no of flip flops

43 (a) SISO

An N-bit SISO register introduce N-clock cycle delay
 App is Digital Delay Line (DDL)

(b) SIPO

used as Serial to parallel converter (or)
 * Temporal to Spatial data converter

(c) PISO

used as Parallel to serial converter
 * Spatial to Temporal data converter

(d) PIPO: used as data storage register.

(44) The main purpose of Johnson counter is to generate unit dist codes

(45) * Binary codes

- (a) Weighted codes : Eg: Binary, BCD
- (b) Non-Weighted codes : Eg: Gray code, Excess-3 code
- (c) Sequential codes : Eg: Binary, BCD
- (d) Self complementing / cyclic codes : Eg: Excess-3, 2421 codes

(46) The only code which is non-weighted & self-complementing is the excess-3

(47) * Expansion of word size

word size $\rightarrow$ No of o/p gates

Let m be the req word size &

n be the available —||—

Then no of chips required is next integer value of $\frac{m}{n}$

(48) * Expansion of word capacity

Let m be the req word capacity

n be the available —||—

Then no of chips req is next integer value of $\frac{m}{n}$

(49) * Saturated logic families are TTL, DCTL, J2L
Unsaturated logic families are TTL schottky, ECL

(50) * TTL logic family

(a) TTL with Totem pole o/p

(b) Open collector configuration.

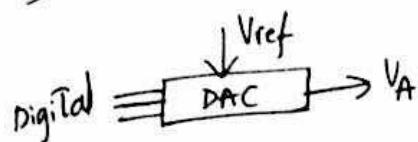
- ⑤1 * In TTL, wired AND logic is possible
In ECL, wired OR logic is possible.

Note: wired AND logic is not possible with TTL-Totem pole configuration ~~but~~ due to current spike problem.
BUT wired AND is possible with open collector configuration

⑤2 DAC { Digital To Analog Circuits }

This is also called as decoding

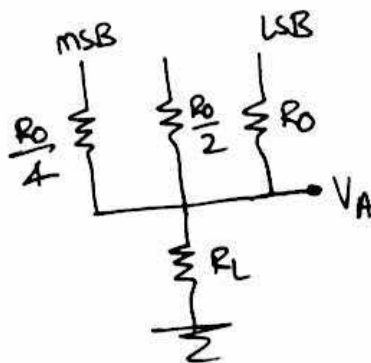
This is required to drive pen recorders, Servo-motors & disc drives



$$V_A \propto \text{digital i/p}$$

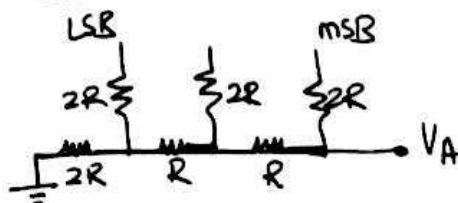
$$V_A = K \times \text{Digital input}$$

⑤3 * Resistive divider network (DAC)



$$V_A = \frac{V_{ref} [\text{Decimal Eq of Digital i/p}]}{2^n - 1}$$

⑤4 * R-2R ladder network (DAC)



$$V_A = \frac{V_{ref} [\text{Decimal Eq of Digital i/p}]}{2^n}$$

* The V_A due to 1st MSB is $V/2$
2nd MSB is $V/4$
3rd MSB is $V/8$ & so on.

⑤⑤ ADC { Analog To Digital Conversion }

This process is also called as encoding

* Flash Type or Comparator method ADC

(a) The no of comparators req is $2^n - 1$
where, $n \rightarrow$ no of o/p bits req

(b) IT takes one conversion cycle to complete the operation

(c) Fastest ADC

⑤⑥ * Successive Approximation Resistive Type ADC (SAR)

IT takes n -conversion cycles if it is an n -bit converter
IT takes equal time for any input signal.

⑤⑦ * Counter type ADC

max conversion time = 2^n clock cycles

Avg ——— = $\frac{2^n}{2} = 2^{n-1}$ clock cycles

IT provides high resolution.

⑤⑧ * DAC performance characteristics

(a) Full scale voltage

$$V_0 = \frac{V_R}{2^n} \text{ for 1 LSB}$$

$$\therefore \text{FSV} = V_R \left[\frac{2^n - 1}{2^n} \right]$$

(b) Resolution: Smallest change that can occur in the o/p when the input is changed by 1 LSB

$$\text{Resolution} = \frac{1}{2^n - 1} = \frac{\text{step size}}{\text{FSV}}$$

58) (C) Accuracy

It is the error between the actual o/p of a DA converter to the expected or theoretical o/p & expressed as a percentage of FSV

If the converter has an accuracy of $\pm 0.1\%$ for a FSV of 10V then the min error for any o/p voltage is $10(0.001) = 10\text{mV}$

Ideally the accuracy should be $\pm \frac{1}{2} \text{ LSB}$ For an 8-bit converter $1 \text{ LSB} = \frac{1}{256} = 0.0039$ or 0.39% of full scale

(d) Settling Time: It is the time taken by DAC o/p to settle within $\pm \frac{1}{2} \text{ LSB}$ of its final value after a change occurs in the i/p

59) ADC

(a) Resolution: is determined by the no of bits chosen to rep the analog quantity

$$1 \text{ LSB} = \text{step voltage} = \frac{\text{voltage range}}{2^n}$$

$$\text{Resolution (V)} = \frac{\text{FSV}}{2^{n-1}}$$

(b) Quantization error: It is the diff b/w digital o/p & analog i/p.

60) FF Timing considerations

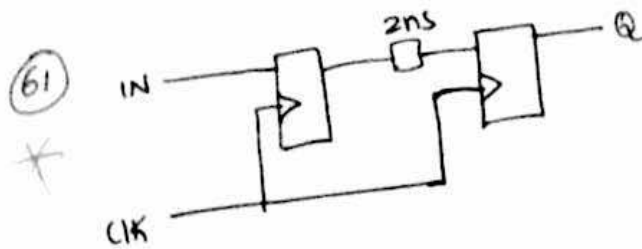
The highest freq of the clock pulse that is to be applied to CLK of FF

$$f_{\text{max}} < \frac{1}{t_s + t_{\text{pd}} + t_{\text{ns}}}$$

$t_s \rightarrow$ setup Time, $t_h \rightarrow$ Hold Time

$t_{\text{pd}} \rightarrow$ Propagation delay of FF

$t_{\text{ns}} \rightarrow$ Propagation delay of next stage.



$$T_{SU} = 0.5ns$$

$$T_H = 0.25ns$$

$$T_{Pdff} = 1.5ns$$

$$f_{max} = \frac{1}{T_{SU} + T_{Pdff} + T_{gate}} = \frac{1}{(0.5 + 1.5 + 2)ns} = 250 MHz$$

data arrival Time = $T_{Pdff} + T_{gate} = (1.5 + 2)ns = 3.5nsec$

* Data require Time (DRT) = $T_{CLK} - T_{SU}$

$$\therefore T_{CLK} - T_{SU} = T_{Pdff} + T_{gate}$$

$$\therefore T_{CLK} = T_{Pdff} + T_{gate} + T_{SU}$$

$$T_{CLK} = 4nsec$$

62 In CMOS

	AND connection	OR connection
p-channel	parallel	Series
n-channel	Series	parallel

63

ADC	Time conversion	ADC	Time conversion
SAR	N clock cycles	counter Type	$(2^n - 1)$ clock cycle
Dual slope	$2^{NH} - 1 \approx 2^{NH}$		
Parallel comparator	1		

64 Dual slope ADC

It is a Tracking ADC
Its has very high accuracy.

SEMICONDUCTOR MEMORIES

Memories can be classified based on

- ① Principle of operation
- ② Based on fabrication technology
- ③ Based on architecture.

① principle of operation

Memories can be classified into

① Sequential access memory

* Eg: CCD { charge Couple Devices }

② Random Access memory

Eg: ROM
RAM

② Based on fabrication tech

- | | | |
|---|-------------------|---------------------|
| ① | Unipolar memories | { MOSFET } Eg: DRAM |
| ② | Bi-polar — — | { BJT & MOSFET } |
| | | Eg: SRAM |

Static RAM

Dynamic RAM

① BJT's & MOSFETs are used

MOSFETs are ^{used} ~~faster~~

② { BJT's are faster than MOSFETs }

Slower

③ 6-Transistors are used to store 1-bit data

2-Transistors are used
{ 1 Transistor, 1 capacitor }
to store 1-bit data

④ Refreshing is not required

Refreshing is required
@ 2msec

③ Based on architecture

(a) Erasable memory Eg: RAM

(b) Non-Erasable memory ROM

(a) Volatile memory Eg: RAM

(b) Non-volatile —||— Eg: ROM

Types of ROM

① ROM

{ Read only memory }

② PROM

{ Programmable ROM }

③ EPROM

{ Erasable PROM }

④ EEPROM [E²PROM]

{ Electrically Erasable PROM }

④ M²NOS Technology

Technology

① Mask Technology

② Electrical fuses

③ UV Tech (FAMOS)

Note: ① FAMOS → Floating Avalanching gate injection Metal Oxide Semiconductor

DIGITAL IC FAMILIES

- ① Propagation delay: is the amount of time lapse during propagation of pulse through a gate from input to output

$$t_{pd} = \frac{t_{PLH} + t_{PHL}}{2}$$

where, t_{PLH} = Delay in going from low to high.
 t_{PHL} = Delay in going from high to low.

② Power dissipated = $VI = \frac{VQ}{T} = V \cdot \frac{CV}{T} = CV^2f$

$$\therefore PD = CV^2f$$

$$\therefore \boxed{PD \propto f}$$

③ $FOM = \text{Speed} \times \text{power dissipation.}$

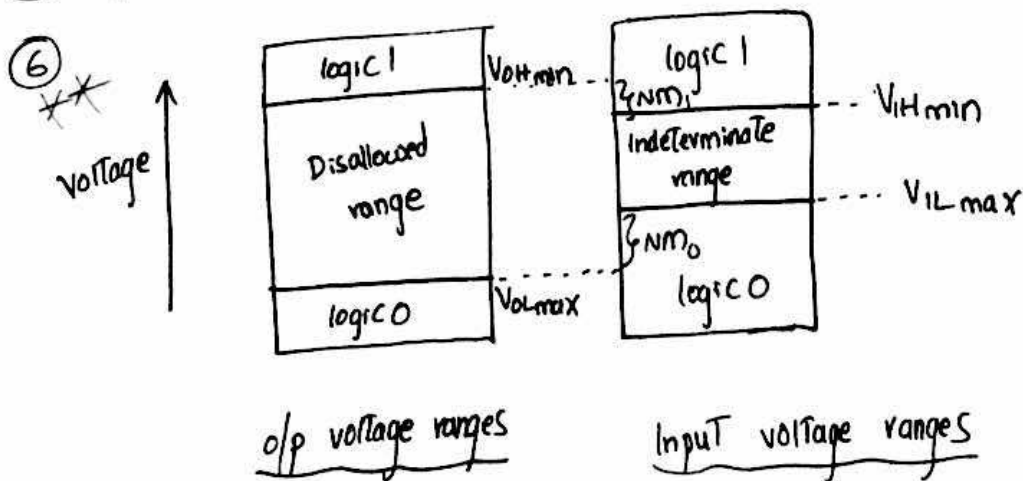
- ① * A 2-bit multiplier can be implemented using
 → 2 input XORs and 4 input AND gates only.
- ② Without any additional circuitry, an 8:1 mux can be used
 To obtain
 → all func of 3 variables & some but not all of 4 variables.
- ③ The 2's complement repr of 8 in 4-bit digital system
 is

- A 1000
 B 0111
 C 1111
 D NOT valid

Ans is (D) NOT Valid
 Since the range of 2's complement
 is -8 to 7

- ④ A switch Tail ring counter is made by using a single
 D-flipflop. The resulting circuit is a T-flipflop.

- ⑤ * An R-S latch is one-bit memory element.



High state noise margin
 $NM_H = V_{OH(min)} - V_{IH(min)}$

Low state noise margin
 $NM_L = V_{IL(max)} - V_{OL(max)}$

final NM
 whichever is smaller
 between these
 two

⑦ * FAN-OUT

The no of inputs that can be driven in low state

$$\text{fan-out}_{(\text{low})} = \frac{I_{OL}(\text{max})}{I_{IL}(\text{max})}$$

The no of inputs that can be driven in high state

$$\text{fan-out}_{(\text{high})} = \frac{I_{OH}(\text{max})}{I_{IH}(\text{max})}$$

TTL o/p acts as ① current sink for low state
② current source for high state.

$$\text{Total fan-out} = \text{fan-out}_{(\text{high})} + \text{fan-out}_{(\text{low})}$$